



PRE-COMPETITION WORKSHOP

FictiPay Datathon Participant **Kickoff** **Guide**

Data description · Problem phases · Evaluation strategy

The FictiPay Universe

350M+

Active Wallets

across Emergland

80M

Daily Transactions

processed & scored

200M+

Transaction Rows

in your dataset

Your challenge: build a complete, production-grade churn prediction pipeline — **not just a model**. Every phase tests a distinct real-world engineering skill.

ACCOUNT TYPES IN THE DATA

- Customer (85%) — Retail wallet users — your churn prediction target
- Merchant (10%) — Businesses accepting wallet payments
- Biller (5%) — Utility companies receiving bill payments

Your Dataset — Three Tables

KYC

ACCOUNT_ID

ACCOUNT_TYPE

ACCOUNT_OPEN_DATE

GENDER

REGION

~2 million rows

One row per account. Small enough to load eagerly into pandas.

Transaction

TrxID

TRX_DATETIME

SRC_ACCOUNT

DST_ACCOUNT

TRX_TYPE

TRX_AMT

200 million+ rows

Too large for RAM. Use Dask / Spark with column pruning.

DayEndBalance

ACCOUNT_ID

DATE

AVAILABLE_BALANCE

~360 million rows (2 M accounts × 180 days)

Largest table. Compute aggregations lazily — never load in full.

The 8-Phase Challenge

Each phase tests a distinct real-world skill · 100 points total · Phase 5 is auto-scored

1

Big Data Handling

10 pts

Dask/Spark · partitioning · sampling

2

Feature Engineering

15 pts

≥10 features · hypotheses · creativity

3

Feature Quality

10 pts

Skew · Pareto · zero-inflation flags

4

Class Imbalance

10 pts

SMOTE · class_weight · undersampling

5

Model Training

20 pts ★ AUTO

LightGBM · LR · AUC / P@K / R@K

6

Hyperparameter Tuning

10 pts

Optuna / Hyperopt · convergence plot

7

Explainability

15 pts

SHAP values · leakage audit

8

Business Recs

10 pts

Decision rule · lift · interventions

OBJECTIVE

Ingest data that doesn't fit in RAM — correctly and reproducibly.

KEY TASKS

- Choose a framework (Dask / Spark) and justify the choice
- Load all three tables with column pruning and partition-awareness
- Document a sampling strategy for safe interactive EDA

EXAMPLE — Dask lazy load

```
import dask.dataframe as dd

trx = dd.read_parquet(
    "transactions/*.parquet",
    columns=["TrxID", "TRX_DATETIME",
            "SRC_ACCOUNT", "TRX_TYPE",
            "TRX_AMT"], # DST_ACCOUNT dropped → -15%
    IO
    engine="pyarrow",
)
# Nothing read yet — Dask builds a lazy graph
print(trx.npartitions) # one partition per month
```

HOW THIS PHASE IS SCORED

framework_justified

/ 2 pts

Rationale vs alternatives (Spark / Polars)

loading_efficiency

/ 4 pts

Column pruning · partitioning · schema

sampling_documented

/ 4 pts

Reproducible strategy · size justified

OBJECTIVE

Craft ≥ 10 behavioral features. Each must have a testable hypothesis about why it predicts churn.

KEY TASKS

- Cover recency, frequency, monetary, balance, type-ratio, and tenure
- Justify every feature — generic rationales score 0
- Deliverable: Markdown/PDF feature catalog

EXAMPLE — Feature catalog (5 of ≥ 10)

Feature	Window	Hypothesis
recency_days	90d	Long gaps signal disengagement
freq_30d	30d	Monthly count captures habit rate
bill_pay_ratio	90d	Bill payments anchor customers
balance_trend	90d	Declining balance = wallet drained
service_diversity	90d	More types = deeper adoption

HOW THIS PHASE IS SCORED

feature_count

/ 5 pts

0 pts <10 · 3 pts 10–14 · 5 pts ≥ 15

feature_rationale

/ 5 pts

Specific behavioral hypothesis per feature

feature_creativity

/ 5 pts

Novel features beyond the problem hints

3 Phase 3 — Feature Quality & Pareto

10 pts

OBJECTIVE

Every feature the model sees must have a well-behaved distribution. Raw financial data is never clean.

KEY TASKS

- Plot distributions; annotate skewness and zero fraction
- Apply log1p, winsorise, or clip to skewed features ($|\text{skew}| > 1$)
- Add indicator flags for zero-inflated features (>20% zeros)

EXAMPLE — Skewness reduction

Feature	Skew raw	Transform	Skew after
freq_7d	+1.37	log1p	-0.03 ✓
TRX_AMT	+4.12	log1p	+0.21 ✓
balance_trend	+6.8	Winsorise	+0.44 ✓
bill_pay_ratio	42% zeros	→ flag	Binary flag

HOW THIS PHASE IS SCORED

distribution_analysis

/ 4 pts

Plots with skew/Pareto annotated

transformations

/ 3 pts

Before/after comparison per feature

sparsity_handling

/ 3 pts

Indicator flags with semantic rationale

4 Phase 4 — Class Imbalance & Sampling

10 pts

OBJECTIVE

~5% churn rate. A model that predicts 'never churn' gets 95% accuracy — and is useless.

KEY TASKS

- Quantify imbalance; explain why accuracy is the wrong metric
- Compare ≥ 2 strategies: SMOTE, undersampling, class_weight
- Justify final choice in terms of precision / recall trade-off

EXAMPLE — Strategy comparison (CV)

Strategy	CV AUC	Prec@10%	Rec@10%
No resampling	0.831	0.312	0.542
class_weight='balanced'	0.863	0.387	0.671
Undersampling	0.847	0.358	0.621
SMOTE	0.851	0.368	0.639

HOW THIS PHASE IS SCORED

prevalence_quantified

/ 2 pts

Rate stated & contextualised

methods_compared

/ 4 pts

≥ 2 methods with metric comparison

approach_justified

/ 4 pts

Tied to FP vs FN business cost

5 Phase 5 — Model Selection & Training

20 pts ★ AUTO-SCORED

OBJECTIVE

Train models that balance performance and scalability. Report correct metrics — not accuracy.

KEY TASKS

- Train ≥ 2 models: e.g. LightGBM and Logistic Regression
- Evaluate on a held-out test set: AUC-ROC, Avg Precision, Prec@K, Recall@K
- Submit predictions.csv — this stage is AUTO-SCORED by the evaluator

EXAMPLE — Model comparison (test set)

Model	AUC	AP	P@10%	Lift
Logistic Regression	0.812	0.324	0.284	3.2×
LightGBM (default)	0.856	0.371	0.371	4.1×
LightGBM (tuned) ★	0.891	0.412	0.412	4.6×

HOW THIS PHASE IS SCORED

AUC-ROC

/ 8 pts

40% weight in composite score

Avg Precision

/ 5 pts

25% weight — handles imbalance

Prec@10%

/ 4 pts

20% weight — top-decile accuracy

Recall@10%

/ 3 pts

15% weight — churn coverage

OBJECTIVE

Improve model performance efficiently. Exhaustive grid search is not acceptable at scale.

KEY TASKS

- Implement Bayesian optimisation (Optuna / Hyperopt) or randomised search
- Show improvement over defaults on the same validation set
- Plot CV score vs iteration and report best parameters

EXAMPLE — Optuna objective

```
import optuna

def objective(trial):
    params = {
        "max_depth" : trial.suggest_int("max_depth", 3,
8),
        "learning_rate": trial.suggest_float(
            "lr", 0.005, 0.3, log=True),
        "num_leaves" : trial.suggest_int("num_leaves",
16, 128),
    }
    return cross_val_score(
        LGBMClassifier(**params, class_weight="balanced"),
        X_train, y_train, cv=5, scoring="roc_auc"
    ).mean()

study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=100)
# Default AUC 0.856 → Tuned AUC 0.891 (+0.035)
```

HOW THIS PHASE IS SCORED

search_implemented

/ 4 pts

Bayesian (4 pts) · Random (2 pts)

improvement_shown

/ 3 pts

Before/after on same validation set

tuning_tracked

/ 3 pts

Convergence chart + best params

OBJECTIVE

A model stakeholders can't understand is a model they won't trust — or use.

KEY TASKS

- Compute SHAP values; produce a beeswarm summary plot
- Explain top drivers in plain business language; flag counter-intuitive ones
- Audit for data leakage — any feature using post-reference data?

EXAMPLE — SHAP feature importance

#	Feature	SHAP	Direction
1	freq_30d	0.423	↓ churn — protective
2	recency_days	0.381	↑ churn — risk signal
3	bill_pay_ratio	0.314	↓ churn — anchors user
4	avg_amt_30d	0.271	↓ churn — high spender
5	std_amt_30d	0.248	↓ churn ← counter-intuitive!

HOW THIS PHASE IS SCORED

shap_provided

/ 5 pts

Beeswarm + dependence plots

insights_discussed

/ 5 pts

Business language · counter-intuitive

leakage_checked

/ 5 pts

Systematic audit · time-split check

8 Phase 8 — Business Recommendations

10 pts

OBJECTIVE

Translate model output into decisions a non-technical executive can act on.

KEY TASKS

- Propose a decision rule (e.g. flag top 10%) with a lift estimate
- Suggest targeted interventions tied directly to SHAP feature insights
- Summarise in a 5-slide deck or dashboard mock-up

EXAMPLE — Lift table

Top %	Customers	Prec	Recall	Lift
5%	3,500	0.38	0.57	7.6×
10%	7,000	0.29	0.87	5.8× ← rule
20%	14,000	0.18	0.72	3.6×

HOW THIS PHASE IS SCORED

decision_rule

/ 3 pts

Threshold + lift + cost frame

interventions

/ 4 pts

Tied to SHAP insights; not generic

presentation_quality

/ 3 pts

Executive-ready · clear · actionable

How Submissions Are Scored

#	Stage	Pts	How	Key evaluation criteria
1	Big Data Handling	10	Human	Framework choice · efficiency · sampling
2	Feature Engineering	15	Human	Count ≥ 10 · rationale · creativity
3	Feature Quality	10	Human	Distributions · transforms · sparsity
4	Class Imbalance	10	Human	Methods compared · justification
5	Model Training	20	AUTO ★	$AUC \times 0.40 + AP \times 0.25 + P@10\% \times 0.20 + R@10\% \times 0.15$
6	Hyperpar. Tuning	10	Human	Search method · improvement · convergence chart
7	Explainability	15	Human	SHAP plots · insights · leakage audit
8	Business Recs	10	Human	Decision rule · interventions · presentation
TOTAL		100		

Key principle: A team with brilliant engineering but a moderate model can outscore a team with a top AUC and no rigour elsewhere.
80 of the 100 points are human-judged.

Submission Checklist

*All files must be present
before your score is computed.*

`predictions.csv`

ACCOUNT_ID + CHURN_PROB $\leftarrow$ AUTO-SCORED

`features.md`

Feature catalog: ≥ 10 features with rationales

`notebook.ipynb`

End-to-end runnable code (or .py pipeline)

`report.pdf`

Model metrics tables + tuning convergence chart

`explainability/`

shap_summary.png + dependence plots

`presentation.pdf`

5-slide executive summary (.ptx accepted)

Good luck · Build something that matters